

19. (3.75 points)

Given the following financial information for an insurer as of December 31, 2016:

Dividend payout ratio:	75.0%
Expected equity market risk premium:	5.0%
Risk-free rate:	2.3%
Insurer's equity beta:	1.10

Calendar Year Projections (\$000)

	<u>2017</u>	<u>2018</u>	<u>2019</u>
GAAP equity (beginning of period)	100,000		
Net income	20,000	22,500	25,000

a. (0.25 point)

Calculate the insurer's risk-adjusted discount rate using the Capital Asset Pricing Model.

b. (1.75 points)

Calculate the value of this insurer as of December 31, 2016, using the dividend discount model.

c. (1.25 points)

Calculate the value of this insurer as of December 31, 2016, using the abnormal earnings model, assuming forecast horizon abnormal earnings are constant after the projected periods.

d. (0.5 point)

Explain the difference in the insurer's calculated value between the dividend discount model and the abnormal earnings method.

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EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 19

TOTAL POINT VALUE: 3.75

LEARNING OBJECTIVE(S): B2

SAMPLE ANSWERS

Part a: 0.25 point

Sample 1

$k = 0.023 + 1.1(0.05) = 0.078$

Sample 2

$k = r_f + \beta(E(r_m) - r_f) = 0.023 + 1.1(0.05) = 0.078$

Part b: 1.75 points

Sample 1

	2017	2018	2019	
Div	20,000(0.75) = 15,000	16,875	18,750	
Beg equity	100,000	105,000	110,625	
ROE = Net income/Beg equity	0.2	0.214	0.226	direct average = 0.213

$g = \rho \times \text{ROE} = 0.25(0.213) = 0.053$

$\Rightarrow V_0 = 15,000/1.078 + 16,875/1.078^2 + 18,750/1.078^3 + 18,750(1.053)/((0.078-0.053)1.078^3)$
 $= 673,828$

Sample 2

Assume NI is after tax.

	2017	2018	2019
NI	20,000	22,500	25,000
Beg Eq	100,000	105,000	110,625
End Cap	105,000	110,625	116,875
Div Paid	15,000	16,875	18,750
ROE	0.200	0.214	0.226

Div Paid = NI $\times$ 75%

End Cap = Beg Eq + NI – Div Paid

Beg Eq = Prior End Cap

ROE = NI/Beg Eq

Since ROE is trending up, select last yr = 0.226

$g = \rho \text{ ROE} = (1-0.75)(0.226) = 0.0565$

$V_0 = \frac{15,000}{1.078} + \frac{16,875}{1.078^2} + \frac{18,750}{1.078^3} + \frac{18,750(1.0565)}{1.078^3} = 778,892k$

Sample 3

	2017	2018	2019
DTV (\$M)	20 $\times$ 0.75 = 15	16.875	18.75
g = Δ Capital / Beg. Equity	20 $\times$ 0.25/100 = 5%	5.625 / 105 = 0.0536	0.0565
BE.	100	105	110.625
End. Equity	100 + 20 $\times$ 0.25 = 105	105 + 5.625 = 110.625	116.875

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Since g is about the same, take an avg $\bar{g} = 5.34\%$

$$V_0 = \frac{15}{1.078} + \frac{16.875}{1.078^2} + \frac{18.75}{1.078^3} + \frac{18.75 \times 1.0534 / (0.078 - 0.0534)}{1.078^3} = 43.403 + 640.919 = 684.322$$

Part c: 1.25 points

Sample 1

	2017	2018	2019
Net Income	20,000	22,500	25,000
Begin Equity (from b.)	100,000	105,000	110,625
AE	12,200	14,310	16,371 = 25,000 - 0.078 * 110,625

$$PV(AE) = 12,200 / (1.078) + 14,310 / (1.078)^2 + 16,371 / (1.078)^3 = 36,700$$

$$PV(TV) = 16371 / (0.078) / (1.078)^3 = 167,542$$

$$V_0 = 100,000 + 36,700 + 167,542 = 304,242$$

Sample 2

Begin Equity 2017	100,000
Begin Equity 2018	100,000 + 0.25 * 20,000 = 105,000
Begin Equity 2019	105,000 + 0.25 * 22,500 = 110,625

$$\text{Abnormal Earnings 2017} = 20,000 - 0.078 * 100,000 = 12,200$$

$$\text{Abnormal Earnings 2018} = 22,500 - 0.078 * 105,000 = 14,310$$

$$\text{Abnormal Earnings 2019} = 25,000 - 0.078 * 110,625 = 16,371.25$$

$$AE = 12,200 / (1.078) + 14,310 / (1.078)^2 + 16,371 / (1.078)^3 + 16371 / (0.078) / (1.078)^3 = 204,244.6$$

$$\text{Value of Insurer} = 100,000 + 204,244.6 = 304,244.6$$

Part d: 0.5 point

Sample 1

Difference is mainly driven by the estimate of the terminal value of dividend (DDM vs AE). Terminal Value DDM is growing at 5.6% after projected period. (Realistically, high growth and high dividend are not sustainable at same time.) Terminal Value AE assumes constant abnormal earning after projection period. (More realistic if firm keeps up with its current operations.)

Sample 2

Abnormal Earnings focuses on value creation and short term forecasts of competitive advantage (this example assumes advantage continues in perpetuity) while DDM just discounts the consequences of value – the dividend. But above the DDM assumes a very high growth rate explaining the higher valuation.

EXAMINER'S REPORT

Candidates were expected to calculate the value of an insurer using the Dividend Discount Model and Abnormal Earnings approaches, and to explain the difference in the results that these two approaches produced.

Part a

Candidates were expected to calculate the insurer's risk-adjusted discount rate using the CAPM.

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Candidates generally performed well on this portion of the question.

Part b

Candidates were expected to calculate the value of the insurer using the Dividend Discount Model. In order to perform the calculation, they needed to consider the information in the question in order to determine an appropriate growth rate to use in the calculation of the Terminal Value.

Common errors include:

- Without sufficient justification, only calculating the ROE/Growth for one year to estimate future Growth, without making the calculations for the other years in the forecast horizon.
- Calculating the Beginning Equities incorrectly, for example by adding the entire Net Income each year.
- Relying on the growth in the actual Dividends during the forecast horizon to estimate the future Growth – this leads to an infinite Terminal Value which is not reasonable.
- Assuming no dividends after the forecast horizon.

Part c

Candidates were expected to calculate the value of the insurer using Abnormal Earnings. Successful Candidates derived Abnormal Earnings and the Terminal Value assuming constant future Abnormal Earnings from 2019 forward then calculated the Value of the Insurer.

Common errors include:

- Making alternative assumptions without sufficient justification about the equity requirements for this part, and did not earn full credit for these assumptions unless a strong justification was provided. Examples of assumptions that did not earn full credit include:
 - Assuming all normal earnings are reinvested without justification.
 - Assuming all abnormal earnings are reinvested without justification.
 - Assuming all net income is reinvested without justification.
- Assuming the growth rate from part b. in the Terminal Value calculation. This assumption is inconsistent with the statement of the question.
- Assuming that the abnormal earnings declined to zero beyond the forecast horizon. This assumption is inconsistent with the statement of the question.
- Forgetting to add the Book Value of the Insurer into the final Value of the Insurer.

Part d

Candidates were expected to explain the difference in the calculated value of the Insurer based on the two methods in part b. and part c.

Common errors include:

- Stating the definitions of the above methods without explaining the difference in the calculated values. For instance, "The dividend discount model values an insurer based on the present value of future dividends. The abnormal earnings method values the insurer based on the PV of future net income above the required return on capital + the beginning book value."
- Failing to explain difference between the results of the two methods in any meaningful way. For instance, "AE focuses on the value created while the DDM focuses on future cash flows to shareholders. Therefore the results in parts b and c are different."